EDUFYI TECH SOLUTIONS

OF INNOVATION

2026 COHORT

GENETIC ENGINEERING

Advanced Curriculum · Virtual Internship

A 12-week, project-driven mentorship in molecular biotechnology — from DNA fundamentals to prime editing, synthetic biology, CAR-T therapeutics and AI-driven genomics — built with industry mentors and six real-time capstone projects.

12 WEEKS	**06** CAPSTONE PROJECTS	**04** CERTIFICATIONS	**20+** LAB TECHNIQUES

WHO WE ARE

Education engineered for employability.

At Edufyi Tech Solutions we transform the academic journey of college students through cutting-edge virtual internships and structured training programs. Every learner is paired with an industry mentor and ships six real-time and capstone projects per domain — the portfolio evidence recruiters actually ask for.

MISSION

Equip students with virtual internships and real-world projects for successful career placement through hands-on industry mentorship.

VISION

Transform education by linking academic learning with practical laboratory and computational experience so students reach their career goals faster.

WHAT MAKES THIS PROGRAM DIFFERENT

Wet-lab + Dry-lab

PCR, cloning and gel work paired with Python, Biopython and NGS pipelines.

Mentor Reviews

Weekly 1:1 code and protocol reviews from practising biotech professionals.

Research Writing

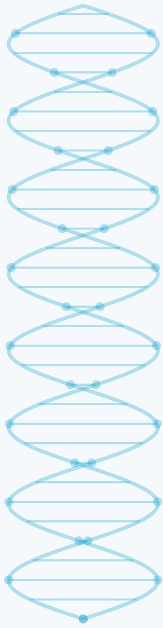
Publish-ready manuscript, poster and patent-landscape training.

Career Support

Resume, portfolio, mock interviews and referral network access.

COURSE TITLE

Genetic Engineering



This course explores the principles and applications of genetic engineering, focusing on the techniques used to read, write and edit genetic material. Learners build a rigorous foundation in molecular biology and bioinformatics, then apply it to therapeutic, agricultural and industrial problems — while engaging seriously with the ethical, legal and social implications of genome modification.

COURSE OBJECTIVES

- Master core genetics, molecular biology and genome architecture.
- Execute the modern editing toolkit: CRISPR-Cas9, base editing, prime editing and cloning.
- Design and deliver gene therapies using viral and lipid-nanoparticle vectors.
- Analyse NGS and single-cell data with Python, Biopython and standard pipelines.
- Apply engineering to medicine, agriculture, industrial biotech and biomanufacturing.
- Evaluate biosafety, biosecurity, IP and regulatory frameworks with evidence.

Toolstack you will actually use

CRISPR-Cas9 / Cas12a · Base & Prime Editors · Gibson Assembly · qPCR & ddPCR · Sanger + Illumina NGS · Benchling · SnapGene · Biopython · Galaxy / GATK · AlphaFold3 · Geneious · FlowJo

WEEK-BY-WEEK OUTLINE

12-week advanced curriculum

Phase 01 — Foundations & the editing toolkit

01 WEEK

Foundations of Genetics & Genomes

- DNA/RNA structure, chromatin and genome architecture
- Mendelian and population genetics, linkage and GWAS basics
- Comparative genomics and genome browsers (Ensembl, UCSC)

02 WEEK

Molecular Biology Techniques

- Replication, transcription, translation and regulation
- PCR, qPCR, ddPCR, gel electrophoresis, blotting
- Nucleic acid extraction, QC and quantification

03 WEEK

Recombinant DNA & Cloning

- Restriction enzymes, ligation, Gibson and Golden Gate assembly
- Plasmid vectors, promoters, selection markers
- Transformation, transfection and electroporation

04 WEEK

CRISPR & Precision Genome Editing

- Cas9, Cas12a, Cas13 mechanisms and guide RNA design
- Base editing, prime editing and epigenome editing (CRISPRa/i)
- Off-target profiling: GUIDE-seq, CIRCLE-seq, mitigation strategies

05 WEEK

Gene & Cell Therapy

- Somatic vs germline; AAV, lentiviral and LNP delivery
- CAR-T and engineered cell therapeutics, allogeneic pipelines
- Case studies: Casgevy, Zolgensma, Luxturna

06 WEEK

Synthetic Biology & Metabolic Engineering

- Genetic circuits, toggle switches, biosensors, BioBricks
- Design-Build-Test-Learn cycles and codon optimisation
- Cell-free systems, minimal genomes and xenobiology

WEEK-BY-WEEK OUTLINE

Applied & frontier modules

Phase 02 — Computation, industry and frontier science

07 WEEK

Bioinformatics & Computational Genomics

- Python + Biopython workflows, sequence alignment, variant calling
- NGS pipelines: QC, mapping, GATK, annotation
- Single-cell RNA-seq, spatial transcriptomics, multi-omics integration

08 WEEK

AI & Protein Engineering

- AlphaFold-class structure prediction and protein design
- Directed evolution and machine-learning-guided variant screening
- Generative models for enzyme and guide-RNA design

09 WEEK

Agricultural & Environmental Biotech

- GMOs, gene-edited crops, drought and pathogen resistance
- Gene drives, biocontainment and ecological risk assessment
- Bioremediation and engineered microbiomes

10 WEEK

Industrial Biotech & Biomanufacturing

- Recombinant protein and mRNA vaccine production
- Bioreactors, upstream/downstream processing, scale-up
- GMP, quality systems and tech-transfer fundamentals

11 WEEK

Bioethics, Biosafety, Law & IP

- Germline editing governance and international frameworks
- Biosafety levels, dual-use research and biosecurity
- Patents, freedom-to-operate and equitable access

12 WEEK

Frontiers & Capstone Delivery

- Organoids, xenotransplantation, synthetic chromosomes
- De-extinction, longevity biology and space biotechnology
- Capstone presentation, peer review and portfolio publishing

SIX REAL-TIME PROJECTS

Build a portfolio, not a certificate collection.

P1

CRISPR Guide-RNA Design Suite

Build a Python tool that designs and scores gRNAs, predicts off-targets and exports a wet-lab-ready protocol.

P2

Plasmid Design & Virtual Cloning

Engineer an expression construct in SnapGene/Benchling, simulate assembly and validate by in-silico digest.

P3

NGS Variant Calling Pipeline

Process raw reads through QC, alignment and GATK to deliver an annotated clinical-style variant report.

P4

Gene Therapy Vector Dossier

Design an AAV/LNP delivery strategy for a monogenic disease with dosing, safety and manufacturing plan.

P5

Synthetic Biology Circuit

Model and specify an inducible genetic circuit or biosensor with a full Design-Build-Test-Learn report.

P6

Single-Cell Atlas Mini-Study

Cluster and annotate a public scRNA-seq dataset and defend the biological narrative you extract.

HOW YOU ARE EVALUATED

Assessment methods

Theory quizzes & module exams	20%
Laboratory & protocol reports	20%
Capstone projects (6)	35%
Research paper / review article	15%
Seminar participation & ethics debate	10%

RECOMMENDED READINGS

- "Molecular Biology of the Cell" — Alberts et al.
- "Genetics: A Conceptual Approach" — Benjamin A. Pierce
- "Molecular Cloning: A Laboratory Manual" — Green & Sambrook
- "A Crack in Creation" — Doudna & Sternberg
- Current literature: Nature Biotechnology, Cell, Nucleic Acids Research

ADDITIONAL RESOURCES

- NCBI, Ensembl, UniProt, PDB and AlphaFold DB access
- Benchling / SnapGene academic workspaces
- Galaxy and Colab notebooks for reproducible analysis
- Wet-lab manuals, SOPs and biosafety checklists
- Recorded mentor masterclasses and journal clubs

Career support included

Interview tricks and most-preferred question banks, resume and cover-letter workshops, LinkedIn and portfolio review, plus mock technical interviews with biotech mentors.

HOW IT WORKS

Our process

01
Schedule a demo call
Discover the programs and how they map to your career goals.

02
Choose a program
Select the plan that fits your goals, timeline and budget.

03
Enroll
Join the cohort community and receive your mentor and lab access.

04
Engage 2–3 months
Commit to weekly sprints, reviews and mentor sessions.

05
Acquire skills & experience
Graduate with six shipped projects and interview readiness.

WHAT YOU TAKE AWAY

Internship Certificate
Verified virtual internship experience letter.

Training Certificate
Confirms completed hours and lab competencies.

Course Certificate
Credential for the full 12-week curriculum.

Letter of Recommendation
Mentor-signed, performance based.

PROFESSIONAL CERTIFICATIONS

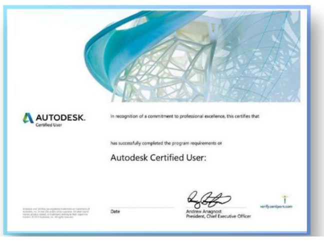
Global MNC certifications.

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.

CISCO
Cybersecurity CCST

MICROSOFT
Azure Fundamentals

IBM
Internet of Things



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ESB
Entrepreneurship

Certifications that travel with your degree

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Authorised testing & certification centre

Certify on campus with globally accepted credentials.

CERTIFICATIONS & OUTCOMES

Certify. Build.
Get hired.

The 12-week genetic engineering track closes with industry credentials, a shipped portfolio and interview-ready evidence of laboratory and computational skill.

- Accredited Certification
- Microsoft Azure Fundamentals
- Cisco Cybersecurity CCST
- IBM Internet of Things
- Adobe Certified Professional
- Autodesk Certified User
- ESB Entrepreneurship
- QuickBooks Certified User

Start your journey

Enrol in the specialised genetic engineering internship and mentorship program with 150+ partnered companies. Learn from industry experts, ship six real-time capstone projects and earn globally recognised certifications.

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UNLOCKING THE POWER OF INNOVATION

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PROGRAM

Genetic Engineering · 12 Weeks

MODE

Virtual · Live Mentorship

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